

Kellogg's® On Location Videos – Eggo® Waffles Video Activity – Teacher Activity Guide

Overview: Students watch a video filmed on location at Kellogg's® to learn how to package and ship Eggo® Waffles.

Materials Needed:

- Eggo® Waffles Video Student Guide - Copies (not front and back, nor collated so that you can hand out the pages one at a time)
- "You're Hired" Signs (don't forget to tag @MathNation & @Kelloggs® when you put these pics on social media)
- Eggo® Box (optional)



Group students (3-4 students per group)

Three Act Tasks -

Three Act Tasks are math activities that consist of three parts.

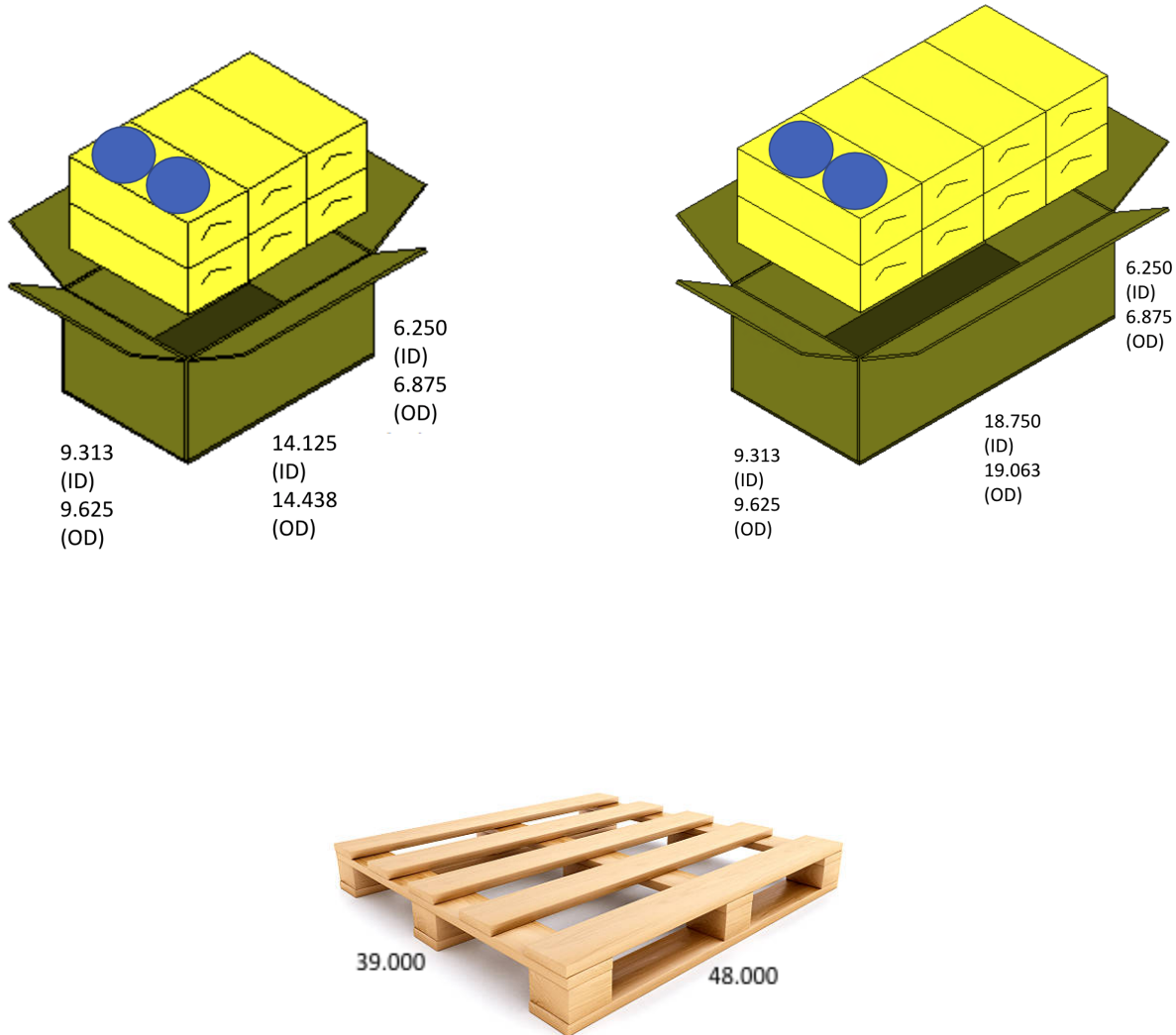
1. The first act is designed to get students engaged in a task that is puzzling. They may not have all of the information necessary to fully answer the question or they may have so many possible answers that it would take days to sort through them all. For that reason, this portion of the task is given a time limit. If students are confused, have them start with: What do you notice? What do you wonder?
2. Act Two swoops in with more information, asking students to commit to one solution. Think of it as a - *Would you rather?* task.
3. The Third Act has students finalize their solutions and share out if they feel that they made the right choice in Task 2. Students do learn to persevere with Three Act Tasks, however, the solutions are revealed allowing students the clarity needed for them to approach the next Three Act with more stamina.

Throughout the tasks, the facilitator should avoid giving any extra information if possible and answer questions with questions. Productive struggle is the name of the game during these tasks!

Task 1: How did you stack up?

With your class, watch the Eggo® Waffles Video. Pause the video, when indicated, and discuss with your group your answer to Task 1. Be prepared to share your answers.

Task: For cost-effectiveness, your group only gets to choose one of the case types 6 or 8 [all measurements are in inches]. Knowing that your stack can't be more than 90" tall, how would your group stack the pallet to get the most (optimize) the truck space?



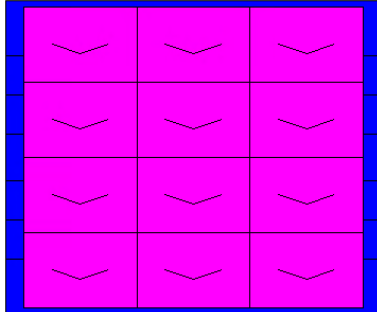
Answers will vary.

The group that stacks the pallet to get the most waffles on the pallet (but keeping it under 90") should be rewarded by being "hired." Take photos of your students with the "You're Hired" sign and post on your school's social media pages! Don't forget to tag Math Nation and Kellogg's®! We can't wait to see you having fun with this activity!

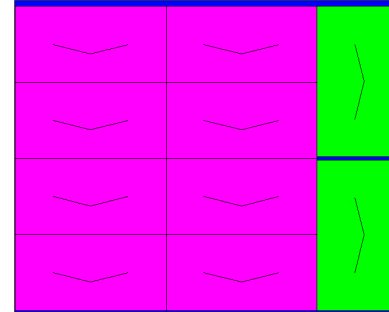
Task 2: Would you rather?

With your class, continue the Eggo® Waffles Video. Pause the video, when indicated, and discuss with your group your answer to Part 2. Be prepared to share your answers.

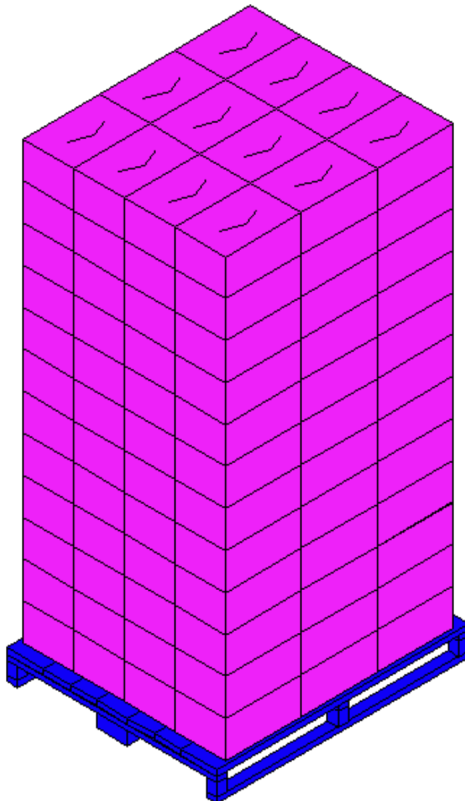
Task: Which stack would your group choose? Be prepared to mathematically defend your answer.



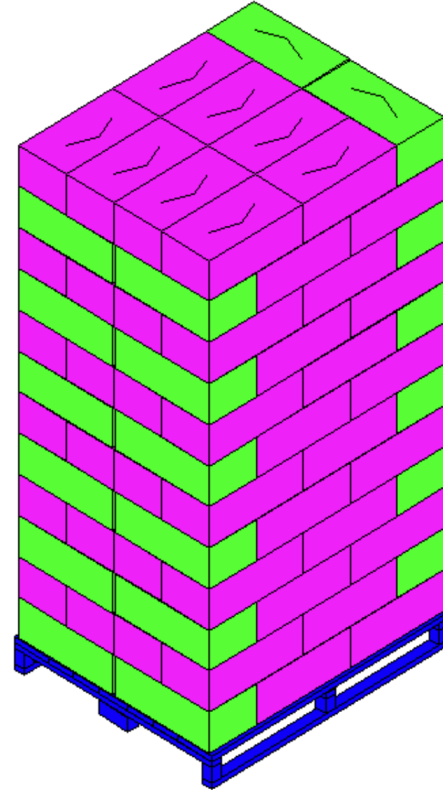
6 Box Case Pallet



8 Box Case Pallet



6 boxes per case
12 cases per layer
12 layers
 $6 * 12 * 12 = \underline{864}$ boxes per pallet



8 boxes per case
10 cases per layer
12 layers
 $8 * 10 * 12 = \underline{960}$ boxes per pallet

Answers will vary.

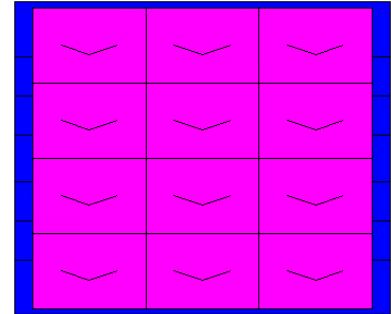
Based on the fact that the 8 box cases can ship 96 more boxes of waffles to be sold, this is the one that Kellogg's® will choose, but Facilitators should be open to all student responses. They may surprise you at their reasoning! Allow students to stick with their answers or switch to the other.

Task 3: Fill it up and Wrap it up!

With your class, watch the Eggo® Waffles Video to the end. Discuss with your group your final task. Be prepared to share your answers.

Task: Danielle explained that 30 of these pallets can fit inside one truck! How much plastic wrap would you need to wrap all 30 pallets just one time (surface area)? How much space will all thirty of those take up in the truck (volume)?

Work Space: (6 box cases)



$$\text{Length} = (14.438 \text{ in} * 3 \text{ boxes}) = 43.314 \text{ in}$$

$$\text{Width} = (9.625 \text{ in} * 4 \text{ boxes}) = 38.500 \text{ in}$$

$$\text{Height} = (6.875 \text{ in} * 12 \text{ layers}) = 82.500 \text{ in}$$

Wrapped Surface Area (30 pallets):

$$\text{SA to be wrapped} = (\text{LW} + 2\text{LH} + 2\text{WH})$$

[only one LW because we only wrap the top not the bottom]

$$\text{SA}_w = (43.314 \text{ in} * 38.500 \text{ in}) + 2(43.314 \text{ in} * 82.500 \text{ in}) + 2(38.500 \text{ in} * 82.500 \text{ in})$$

$$\text{SA}_w = 15,166.899 \text{ in}^2$$

We need the Surface Area of 30 pallets so...

$$30 \text{ SA}_w = 455,006.970 \text{ in}^2$$

[rarely do we think in inches squared when thinking about plastic wrap so we must divide by 144 in² to convert to ft²].

$$\text{SA}_w / 144 \approx 3,159.771 \text{ ft}^2$$

Volume (30 pallets):

$$V = \text{LWH}$$

$$V = 43.314 \text{ in} * 38.500 \text{ in} * 82.500 \text{ in}$$

$$V = 137,576.0925 \text{ in}^3$$

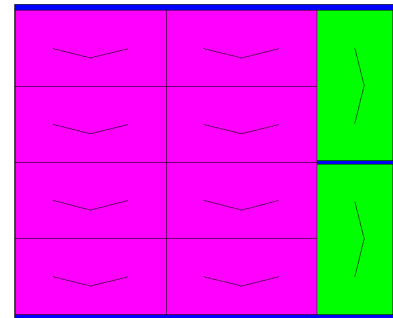
$$30 V = 4,127,282.775 \text{ in}^3$$

[divide by 1728 in³ to convert to ft³]

$$V \approx 2,388.474 \text{ ft}^3$$

Task: Danielle explained that 30 of these pallets can fit inside one truck! How much plastic wrap would you need to wrap all 30 pallets just one time (surface area)? How much space will all thirty of those take up in the truck (volume)?

Work Space: (8 box cases)



Length = $(19.063 \text{ in} * 2 + 9.625 \text{ in}) = 47.751 \text{ in}$
[2 of the long sides and one of the short side]

Width = $(9.625 \text{ in} * 4 \text{ boxes}) = 38.500 \text{ in}$
[4 short sides is longer than 2 long sides! see the gap?]

Height = $(6.875 \text{ in} * 12 \text{ layers}) = 82.500 \text{ in}$

Wrapped Surface Area (30 pallets):

SA to be wrapped = $(LW + 2LH + 2WH)$
[only one LW because we only wrap the top not the bottom]

$SA_w = (47.751 \text{ in} * 38.500 \text{ in}) + 2(47.751 \text{ in} * 82.500 \text{ in}) + 2(38.500 \text{ in} * 82.500 \text{ in})$
 $SA_w = 16,069.8285 \text{ in}^2$

We need the Surface Area of 30 pallets so...

$30 SA_w = 482,094.855 \text{ in}^2$

[rarely do we think in inches squared when thinking about plastic wrap so we must divide by 144 in^2 to convert to ft^2].

$SA_w / 144$
 $\approx 3,347.881 \text{ ft}^2$

Volume (30 pallets):

$V = LWH$
 $V = 47.751 \text{ in} * 38.500 \text{ in} * 82.500 \text{ in}$
 $V = 151,669.1138 \text{ in}^3$
 $30 V = 4,550,073.413 \text{ in}^3$

[divide by 1728 in^3 to convert to ft^3]

$V \approx 2,633.144 \text{ ft}^3$

Activity Synthesis:

Facilitators should have groups mathematically defend their answers to the class. Once again you should use your “hiring” sign to welcome the group with the best answer to the Kellogg’s® Family!

Extensions:

- Based on the picture on the first page of the activity there are 6 waffles in each box.
 - How many total waffles are in each pallet?
 - In a truckload (30 pallets)?
- Estimate the amount of syrup that can fit into the holes of an Eggo® Waffle.
 - Use measurement to determine the size of the waffle holes.
 - Find the volume of each hole/entire waffle.
 - Convert from cubic cm (ml) to fl oz.
- Design a cylindrical container to package the Eggo® Waffles (think Pringles® can)
 - Use measurement to determine the diameter and height of the waffle.
 - Calculate the circumference of the waffle.
 - Determine how much cardboard and/or plastic for end caps would be needed to create the box (surface area).
 - Compare to the surface area of the current box, is it worth the money/effort?
 - Determine the volume of the new Eggo® Container.

You're Hired!



Kellogg's®

